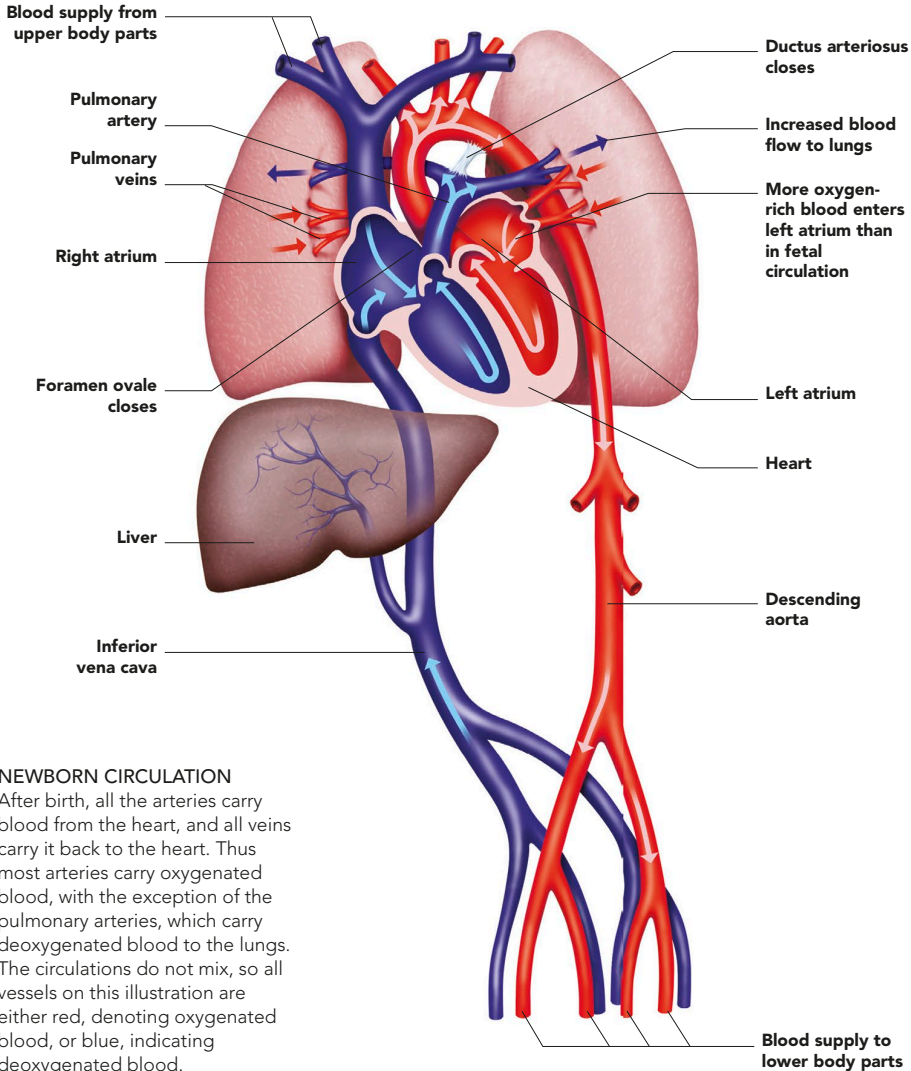


CIRCULATION AT BIRTH

At birth, the baby takes its first breaths and the umbilical cord is clamped. This forces the circulatory system into a monumental response: to convert itself immediately to obtain its oxygen supply via the lungs. Blood is sent to the lungs to retrieve oxygen, and the pressure of this blood

returning from the lungs into the left atrium forces shut the foramen ovale between the two atria, thus establishing normal circulation. The ductus arteriosus, the ductus venosus, and the umbilical vein and arteries close up and become ligaments.



NEWBORN CIRCULATION

After birth, all the arteries carry blood from the heart, and all veins carry it back to the heart. Thus most arteries carry oxygenated blood, with the exception of the pulmonary arteries, which carry deoxygenated blood to the lungs. The circulations do not mix, so all vessels on this illustration are either red, denoting oxygenated blood, or blue, indicating deoxygenated blood.